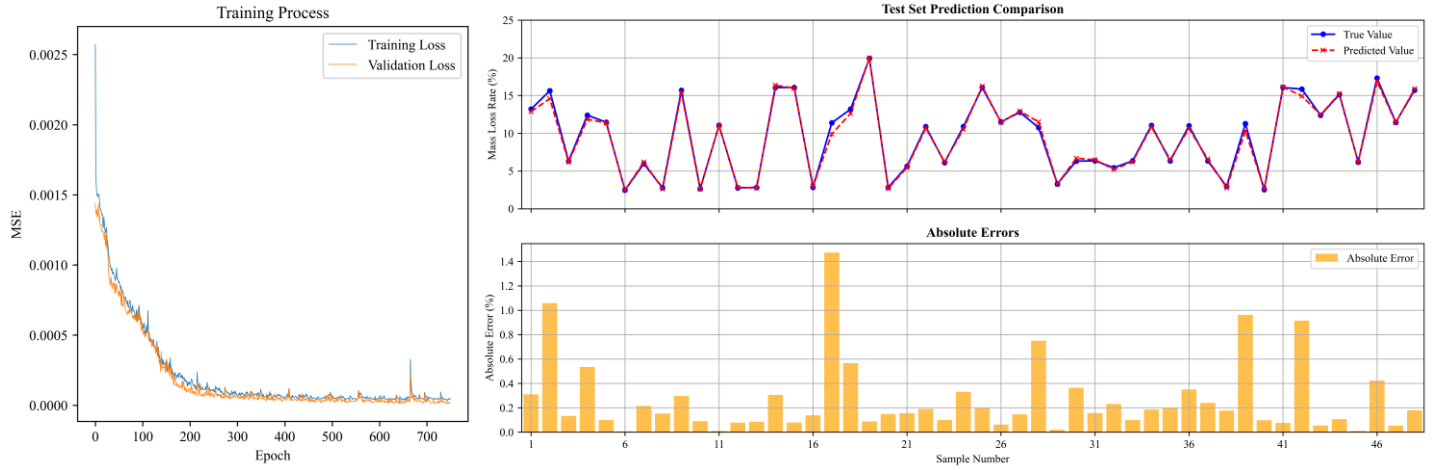
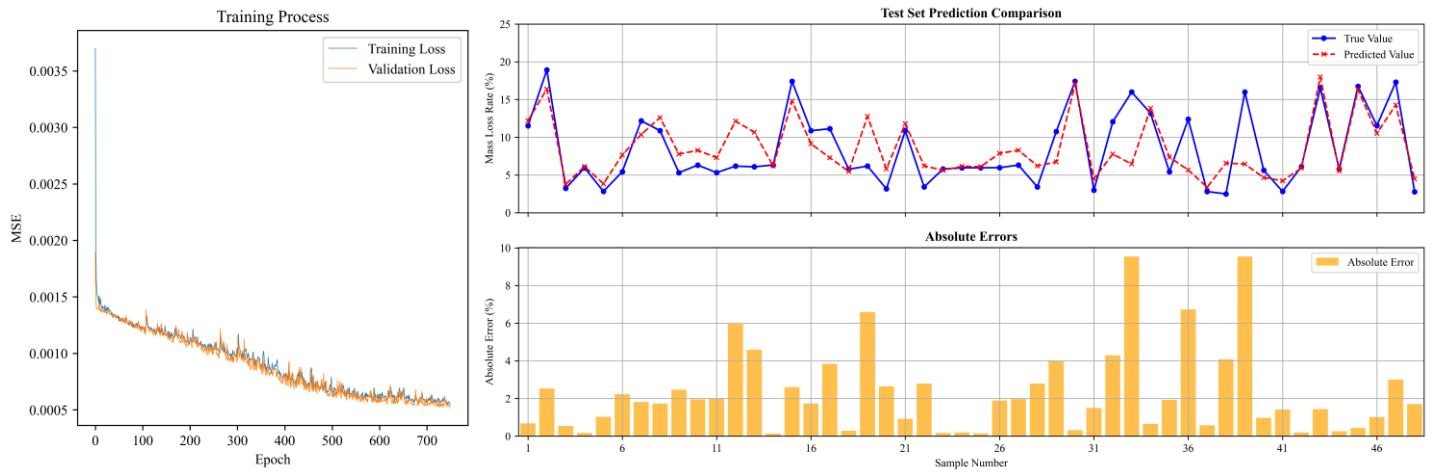


Prediction Results with Attention (The model I developed)



Prediction Results with Attention Removed



Model	Absolute Error Range	Average Absolute Error	Relative Error Range	Average Relative Error
With Attention	[0.0001, 0.0147]	0.0026	[0.08%, 12.94%]	2.98%
Attention Removed	[0.0014, 0.0954]	0.0229	[1.84%, 163.85%]	34.10%

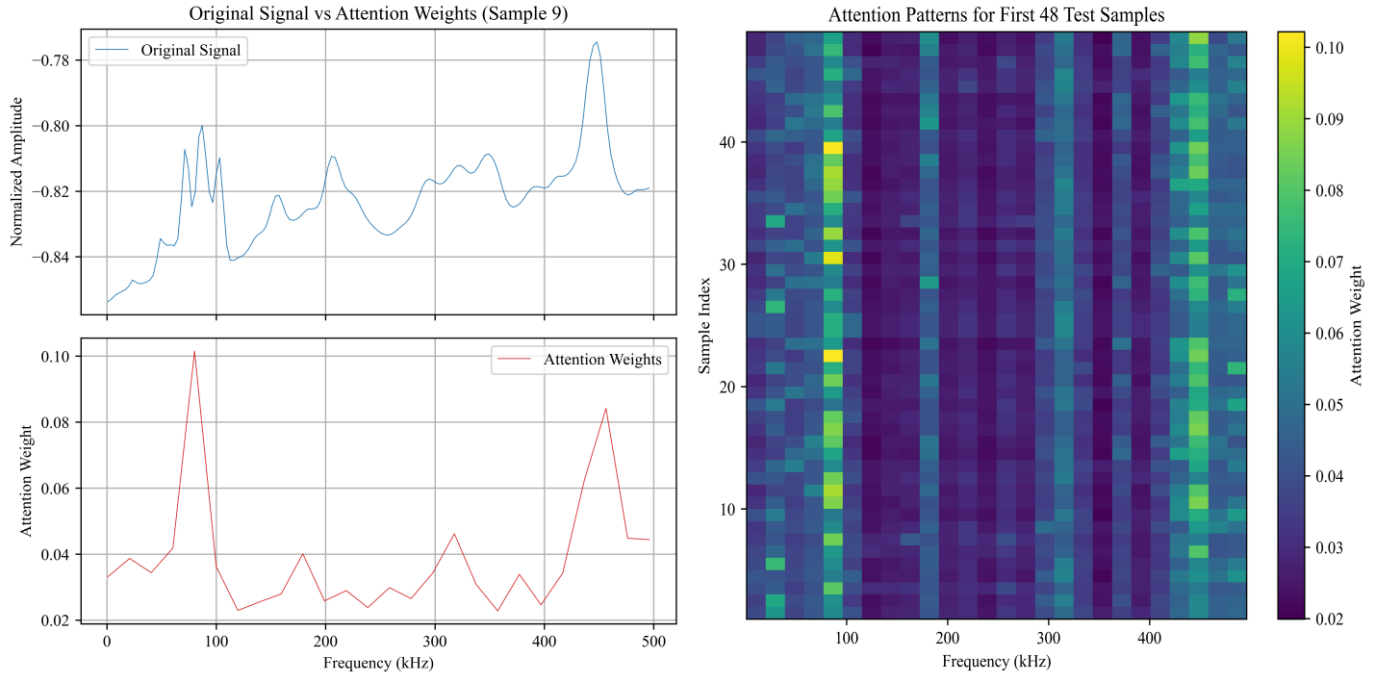
Attention Mechanism under Noisy Impedance Signals

This implementation investigates the effectiveness of an **attention mechanism for corrosion quantification from noisy piezoelectric impedance signals**. When noise is introduced, different frequency regions may contain substantially different levels of useful information. The attention mechanism therefore enables the model to **automatically assign different weights to different frequency segments**, emphasizing informative regions while reducing the influence of less informative or noise-dominated components.

The introduction of attention resulted in a **significant improvement in prediction accuracy**, as demonstrated by the comparison below.

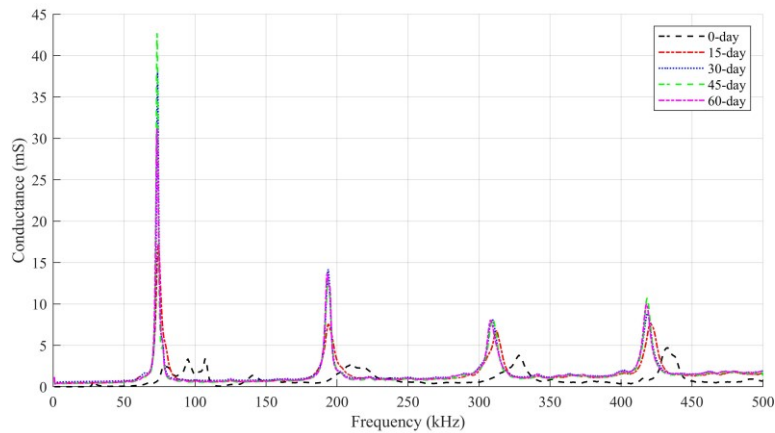
Attention Weight Analysis

The visualization of attention weights demonstrates that the model successfully **learns to automatically distribute attention across different frequency regions** rather than treating all signal segments equally.

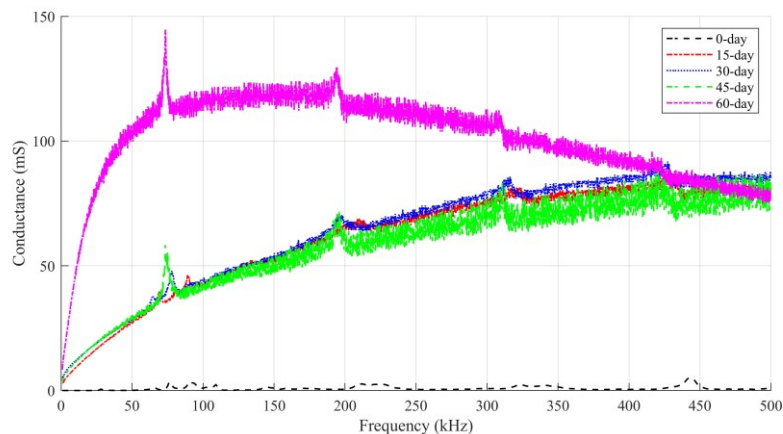


To generalize, these results suggest that different segments within a signal may contain varying levels of information. From my perspective, this indicates that attention mechanisms may provide a useful way to identify and emphasize potentially more informative signal regions, which could contribute to improved prediction accuracy, particularly under noisy conditions.

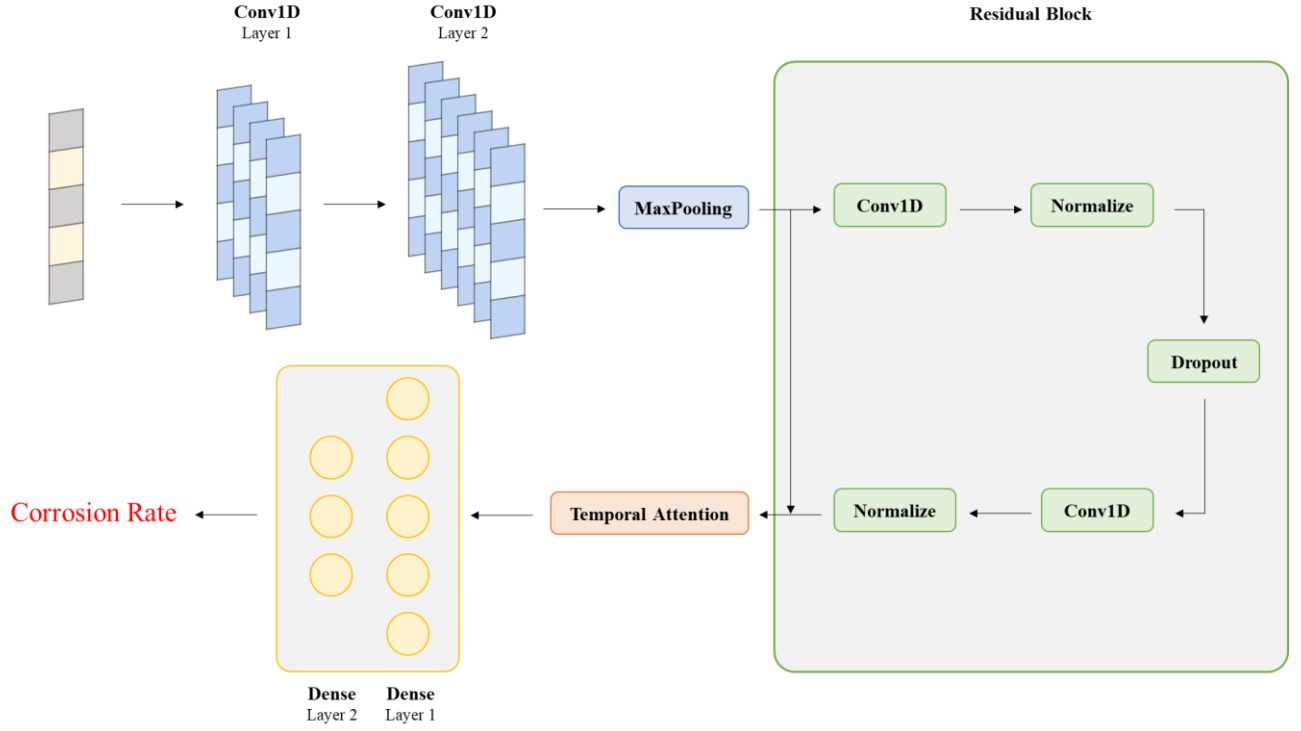
Normal Signals



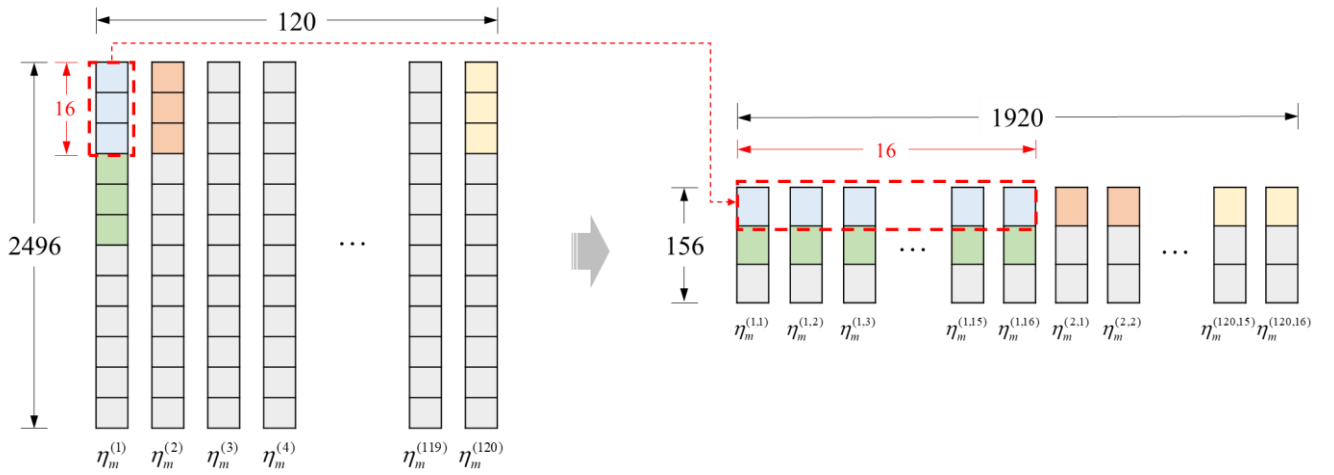
Noisy Signals



The Model Architecture I developed



Data Augmentation



Attention Mechanism

